

Create azure quota alerts tool

Wednesday, September 11, 2024 11:34 AM



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Description of Azure PowerShell Script for Quota Alerts

This Azure PowerShell script automates the creation and management of quota alerts in a specified Azure subscription. The script performs a series of tasks to ensure that the necessary infrastructure is in place and that alerts are configured correctly.

Key Functions and Steps

Pre-setup

- **Set Maximum Variable and Function Count:** Limits for variables and functions are set to ensure the script runs smoothly.
- **Connect to Azure Account:** The script connects to the Azure account using managed identity for authentication.
- **Suppress Azure PowerShell Breaking Change Warnings:** Suppresses warnings about breaking changes in Azure PowerShell to avoid disruptions.

Setup Storage Account and Resource Group

- **Select Subscription:** The script sets the subscription in which resources will be managed.
- **Define Storage Account Information:** Specifies the names and regions for storage accounts and containers used for storing templates and alert configurations.
- **Set Azure Context:** Sets the Azure context to the specified subscription and tenant.
- **Create Resource Group if it doesn't exist:** Checks if the specified resource group exists; if not, it creates a new one with the provided tags and location.
- **Create Storage Account if it doesn't exist:** Checks for the existence of the specified storage account; if not found, it creates a new storage account with specified configurations.
- **Create Storage Containers if they don't exist:** Ensures the specified storage containers are present in the storage account; creates them if they do not exist.
- **Set Storage Context:** Retrieves storage account keys and sets the storage context for subsequent operations.

Create or Select Resource Group

- **Function to Create New Resource Group:** A function (NEWRGCREATE) is defined to create a new resource group based on user input and selected region.
- **Select Subscription for Alerts and Action Group:** Prompts the user to select a subscription for setting up alerts and action groups.
- **Get Region/Location for Request:** Prompts the user to select a location from available Azure regions.
- **Get SKU Family:** Retrieves and prompts the user to select a SKU family quota to add an alert to.

Create Action Group

- **Prepare Email Receivers:** Identifies users and groups with specific roles (Owner, Contributor) in the selected subscription. Collects email addresses of these users and groups to add them to the action group for receiving quota alerts.

Final Steps

- **Create or Update Action Group:** Checks if an action group exists; if not, creates a new one and adds the collected email addresses. If the action group exists, updates it with any new email addresses not already included.
- **Generate and Deploy Alert Template:** Reads existing alert templates and parameters from storage. Modifies the template to include user-specified alert names, action groups, and scopes. Deploys the updated template to the selected resource group, creating the quota alert.

Summary

This script ensures that the necessary infrastructure (resource groups, storage accounts, containers) is in place, collects relevant user and group information, and sets up quota alerts in Azure. It automates the process of creating and managing quota alerts, making it easier to monitor and manage Azure resources efficiently.